

Vendor Comparison and Procurement Assistant

Summary

A working assistant that reads three inconsistently formatted vendor quotations (PDF, two currencies, bundled vs. itemised pricing), standardises them, scores each vendor on five weighted criteria, checks every quote against procurement policy and proposes a recommendation.

A procurement officer reviews the result in a live dashboard, re-weights the criteria, records an approve / override / clarify / reject decision with a justification, and exports the audit trail. Claude drafts the approver summary, clarification e-mails to flagged vendors, negotiation points and answers policy questions. The system never places orders or commits funds.

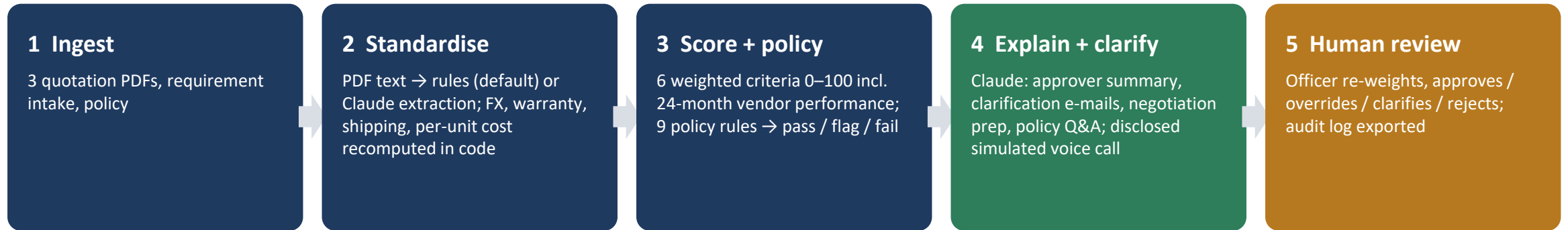
Problem / opportunity

Fictional scenario: Bramwell & Co. Consulting must buy 40 laptops for new hires (USD 60k budget, 21-day deadline). Every purchase over USD 25k needs three quotes, but each vendor answers differently: EUR vs USD, warranty extensions priced separately, shipping bundled or not, deposits, short validity.

Comparing them by hand takes an analyst an afternoon, and policy breaches (advance payments, missing ISO 27001) are often found by Finance after the PO. The opportunity: cut comparison time from ~4 hours to under 45 minutes and catch every policy exception before approval, while keeping the decision with an accountable human.

Solution overview

Deterministic code calculates, generative AI explains and drafts, a human decides.



Generative AI is used for

- Reading messy quote text into a strict JSON schema (optional mode)
- Plain-language recommendation summary for the approver
- Clarification e-mails and a disclosed, simulated voice call to flagged vendors
- Negotiation talking points (no competitor prices disclosed)
- Natural-language Q&A on the procurement policy, citing rule ids

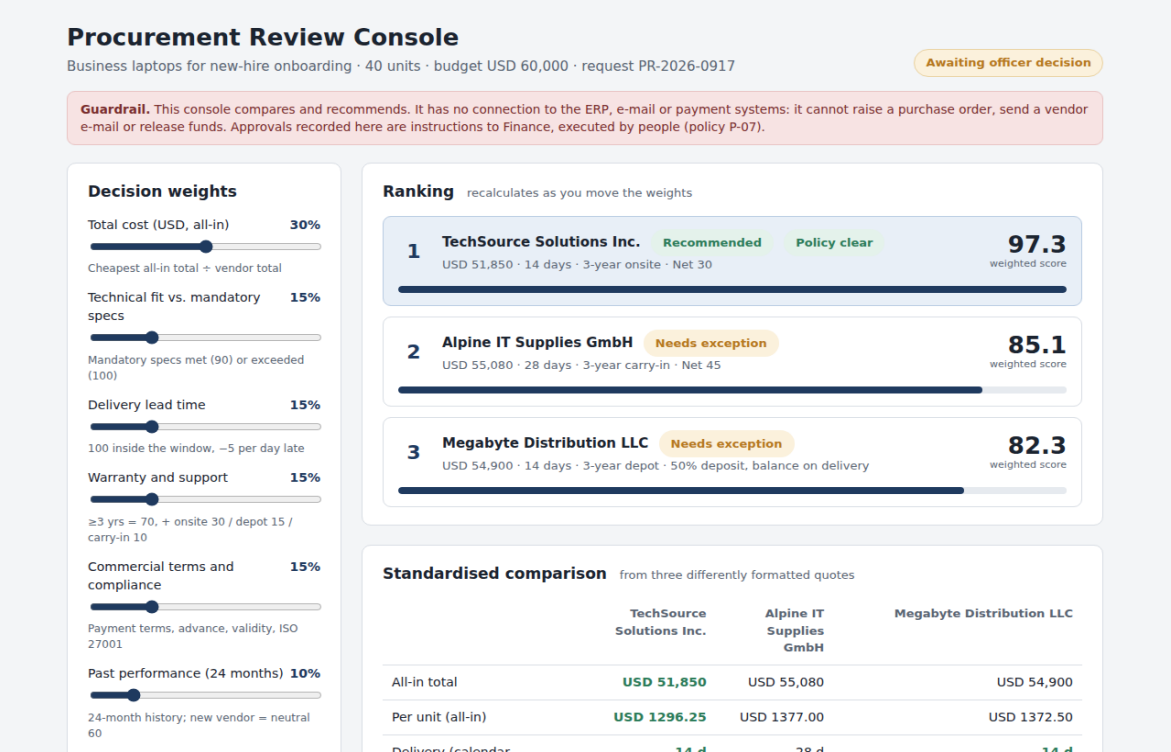
Deterministic logic is used for

- Currency conversion, totals, per-unit cost
- All five criterion scores (formulas printed in the report)
- Weighted ranking that recalculates live in the dashboard
- Policy checks: budget, payment, warranty, ISO 27001, validity, vendor history
- Past-performance score from 24 months of order data
- Eligibility: hard fails are excluded from the recommendation

Key deliverables (Project Files)

- src/ — 5 numbered Python scripts + run_pipeline.py
- data/ — intake, policy, 3 fictional quotation PDFs
- outputs/ — standardised data, comparison report, AI documents, voice call audio + transcript, audit log, screenshots
- app/dashboard.html — self-contained review console
- docs/ — Requirements_Coverage, Demo_Script, README with metrics and limitations

Result and the human decision point



Ranking with default weights — recommendation highlighted, policy status per vendor

97.3

TechSource — recommended, all policy checks pass

85.1

Alpine — ISO 27001 pending, 28-day delivery

82.3

MegaByte — 50% deposit, new vendor; voice call obtained 30% option

Officer decision

required before anything moves forward

Procurement officer

Priya Raman

Decision

Approve the recommended vendor

Justification (required when not approving the recommendation)

Why this decision? This text is stored in the approval history.

Record decision

Recorded.

Finance can now raise the PO for TechSource Solutions Inc. — the assistant does not do this.

Approval history

audit trail of every decision recorded in this session

Time	Officer	Decision	Vendor	Weights	Justification
2026-09-10 16:46	Priya Raman	Approved recommendation	TechSource Solutions Inc.	30/15/15/15/15/10	Lowest cost, only fully compliant bidder, delivery inside onboarding window.

Officer records an attributable decision; the audit trail stores weights, scores and justification

No autonomous purchasing: no e-mail, ERP or payment code; e-mails are drafts; the voice agent discloses it is an AI and refuses to commit; approval is a JSON instruction executed by Finance (P-07, P-09).

Business metrics to track: analyst time per three-quote comparison (~4 h → < 45 min) and share of policy exceptions caught before PO (target 100%).

Limitations: parser tuned to demo layouts; linear scoring; fixed FX; simulated voice; AI text can be fluent but wrong, so every AI document is marked for human review. Before deployment: pilot on 10 historic purchases, SSO, immutable audit log, ERP integration, telephony with consent capture, CISO and Finance sign-off.